# JAFFEDRONE

**A drone instrument and meditation tool for the Miyoo Mini Plus**

User manual · version 0.10.0




github.com/sydilaxe/Jaffedrone-dev

# Getting started




Jaffedrone is a drone of five voices: strings that are bowed and plucked, or wind voices that are blown and tongued, tuned to a scale and moving through chords. Each voice has its own rhythm, or you play it by hand.

## Install

Unzip Jaffedrone-miyoo.zip at the root of the SD card (it holds App/Jaffedrone/) and open Apps → Jaffedrone. The start-up screen shows for a moment; any button moves on.

## The first buttons

| | | |
|---|---|---|
| 1 | START | start or stop the drone (it fades in and out) |
| 2 | START + L1 / R1 | previous / next page: PLAY, SOUND, COMPOSE, MIX, SONG |
| | SELECT | the next chord |
| 3 | MENU | leave the app |

Listen on headphones for the deepest low end and for binaural beats; the speaker plays little below about 200 Hz.

## The five pages



PLAY: the strings and their rhythms



SOUND: each voice's tone



COMPOSE: scale, chords, change



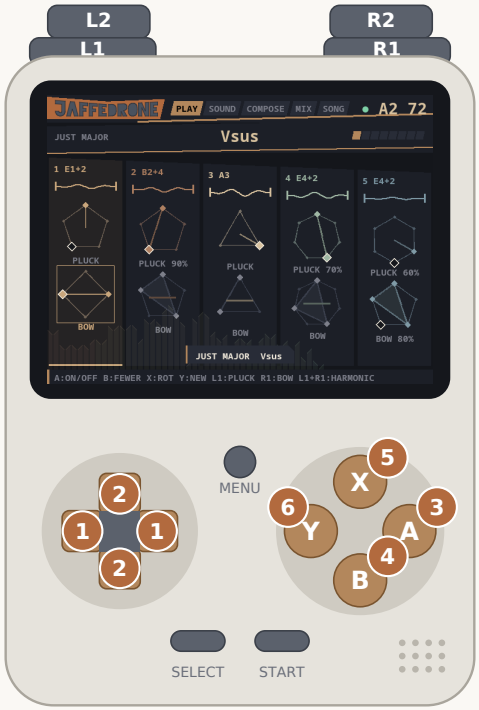
MIX: levels and effects



SONG: presets, chord memory, key

START + R1 moves right along the tabs at the top, START + L1 left. On SOUND, COMPOSE and MIX, L1 and R1 alone change page too.

# PLAY: the rhythm rings




Each string has two Euclidean rings: the pluck ring plucks on its pulses and the bow ring swells the bow on its pulses (off, the bow plays steadily). A ring spreads its pulses as evenly as it can over its steps, like the rhythms of HI5 and RadioScan.

| | | |
|---|---|---|
| 1 | ◄► | choose a string |
| 2 | ▲▼ | choose its pluck ring or bow ring |
| 3 | A | switch the ring on or off |
| 4 | B | fewer pulses, wrapping back to all |
| 5 | X | rotate the pattern |
| 6 | Y | a new simple pattern |

A label like PLUCK 70% means each pulse of that ring plays seven times in ten (set with R2 + X / Y).

## Shaping a ring: hold L2




| | | |
|---|---|---|
| 1 | hold L2 | then: |
| 2 | ▲▼ | more / fewer steps |
| 3 | ◄► | step length (right is faster): 1/16 to 2 bars |
| 4 | X | more pulses |
| 5 | Y | rotate back |
| 6 | A | mute or unmute the string (a red bar marks it) |
| | tap L2 | the quiet view (see The quiet view) |

# Playing by hand

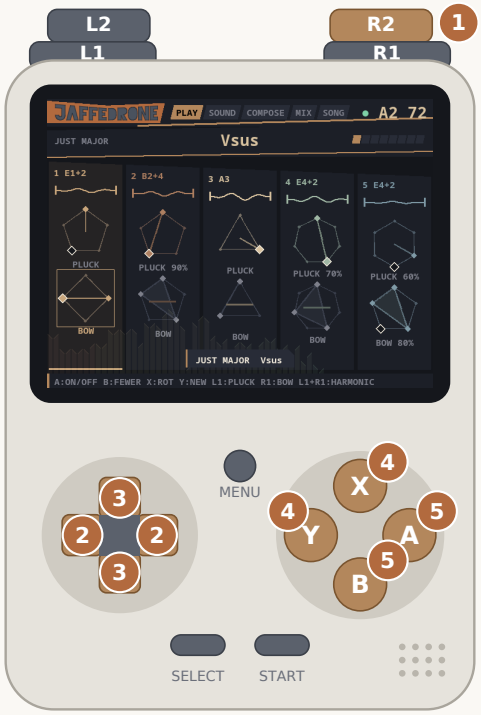
The shoulder buttons play the string selected on PLAY, over whatever its rings are doing. A wind voice is tongued and blown instead.




| | | |
|---|---|---|
| **1** | tap L1 | strike the selected string like a hammered dulcimer: a bright hammer hit that bounces (wind: a hard attack that overblows for a moment and settles) |
| | hold L1 briefly | pluck it (wind: tongue a note); a tap is under about a tenth of a second |
| **2** | R1 held | bow it: the bow is drawn in slowly, swelling over about two seconds and brightening as it bites, and sings while you hold (wind: blow) |
| | L1 + R1, either first | bow with a light touch at a node: the next harmonic up sings (wind: overblow, an octave up, a twelfth for the reed). Bowing with R1, add L1 for overblows as you bow |

The bowed string's column lights while the bow is on it.

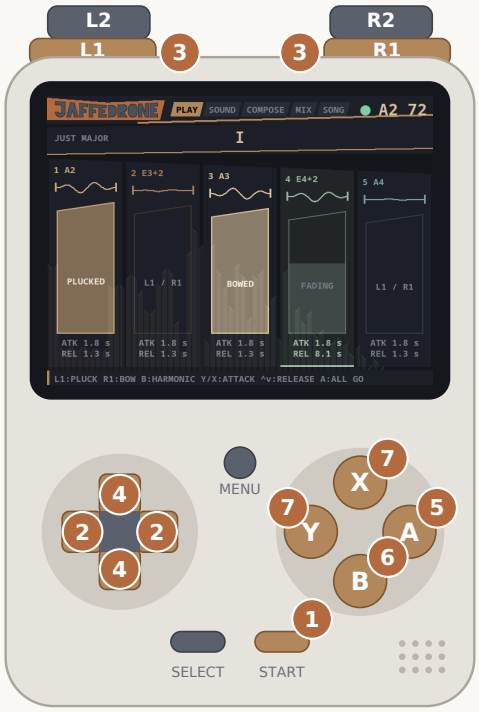
## Quick sound changes: hold R2




| | | |
|---|---|---|
| **1** | hold R2 | then: |
| **2** | ◄ ► | previous / next scale |
| **3** | ▲ ▼ | tone: brighter / darker |
| **4** | X / Y | the selected ring's chance, more / less |
| **5** | A / B | body resonance, more / less |

# LATCH: hold each string yourself

In LATCH the rings rest and every string is silent until you play it. A string you pluck or bow keeps droning, with no need to hold the button, until you play it again; then it fades over its own RELEASE.




| | | |
|---|---|---|
| 1 | START + A | switch PLAY MODE between RINGS and LATCH (also on the SONG page) |
| 2 | ◄ ► | choose a string |
| 3 | L1 / R1 | pluck (a tap strikes) / bow it on: it drones. Either again: it lets go and fades |
| 4 | ▲ ▼ | that string's RELEASE, 0.05 s to 20 s |
| 5 | A | let every string go |
| 6 | B held, or L1 + R1 | the next harmonic up on the droning string (wind: overblow); with both shoulders, either first, it lasts while both are held |
| 7 | Y / X | that string's bow ATTACK, shorter / longer, 0.05 s to 10 s |

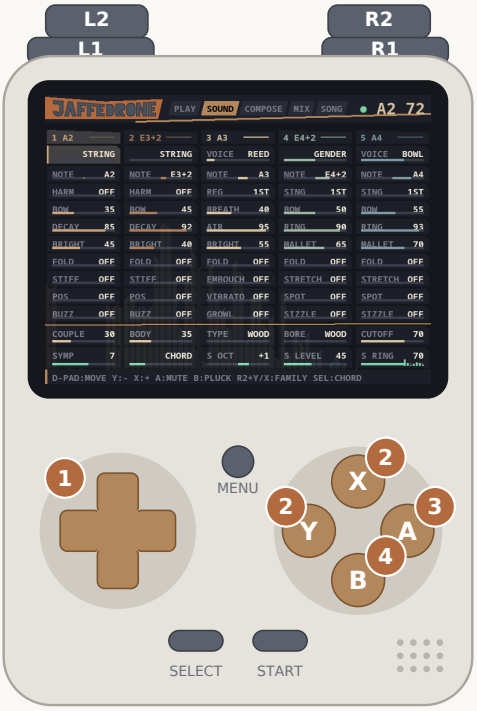
Each column shows a board with a shed-roof top. It fills while its string drones and drains as it fades, marked STRUCK, PLUCKED, BOWED, HARMONIC or FADING, with its ATTACK and RELEASE beneath. A pluck or strike starts at once; a bow swells in over its ATTACK. RELEASE is also on the MIX page; LATCH, ATTACK and RELEASE are saved with songs and presets.

## Try it

Press START + A. Bow string 1 with R1 for a low drone, move to string 3 and pluck it with L1, then press L1 on string 3 again to let it ring away. Raise string 5's RELEASE with ▲ to ten seconds or more for a long dying shimmer. Tap L1 on string 4 to strike it, and hold B to let its harmonic sing. Lower string 1's ATTACK with Y for a bow that bites at once.

# SOUND, COMPOSE and MIX

On these pages the d-pad moves to the nearest control in that direction, and Y / X lower / raise it. Held, they speed up.




| | | |
|---|---|---|
| 1 | d-pad | move between controls |
| 2 | Y / X | lower / raise the control |
| 3 | A | on a string's control: mute or unmute that string |
| 4 | B | pluck the string (to hear a change) |

## SOUND

The VOICE row makes each column a STRING, one of seven winds (LIP, REED, DOUBLE, FLUTE, FREE, RECORDER, BANSURI) or one of nine struck voices (see Struck voices). R2 + Y / X on VOICE jumps to the previous / next family; Y / X alone step one at a time. The rows below follow the voice: a string has BOW, DECAY, BRIGHT, FOLD, STIFF, POS and BUZZ; a wind BREATH, AIR, EMBOUCH, VIBRATO and GROWL; a struck voice BOW, RING, MALLET, STRETCH, SPOT, SIZZLE and SING. The shared row holds COUPLE, BODY, its TYPE, the winds' BORE and CUTOFF; the last row the sympathetic strings.




COMPOSE: the scale and root (a wheel of the scale's notes, the chord lit), how often chords change and how: MOTION, HOME, COLOR, CHANCE, RELATE, EVOLVE, GLIDE, tempo, Wi-Fi sync and two LFOs.




MIX: each voice's LEVEL, PAN and GATE (RELEASE in LATCH), the master, undertones (SUB), then delay, reverb and binaural.

# Struck voices

A struck voice is a physical model of a vibrating body: its resonant modes, struck by a mallet (a mass on a spring that meets the surface and bounces off, so a hard mallet makes a short bright contact and a soft one a long dark push) or bowed (stick-slip friction, so the modes it meets grow until they sing, as on a rubbed bowl or a bowed cymbal). Every voice can be struck and bowed.




| | | |
|---|---|---|
| 1 | L1 | strike (a tap: a hard mallet, hit harder) |
| 2 | R1 | bow: the body catches and sings |
| 3 | L1 + R1, B | a touch: the fundamental is damped and the next mode rings and sings |

| | |
|---|---|
| **GENDER** | gamelan metallophone bars, in beating pairs (ombak) |
| **BONANG** | gamelan kettle gongs, paired |
| **GONG** | a large gong, low, its pairs beating slowly |
| **BOWL** | a singing bowl: struck, or rubbed until it sings |
| **HANDPAN** | fundamental, octave and fifth above, tuned |
| **STEELPAN** | the same family, brighter |
| **TONGUE** | a steel tongue drum: pure, a bar held at one end |
| **CYMBAL** | a dense shimmer of modes; bowed, it screams and sings |
| **DRUM** | a membrane: a short pitched thud |
| **RING** | how long it rings |
| **MALLET** | soft felt to hard metal |
| **STRETCH** | pulls the modes apart: more clangor, more metal |
| **SPOT** | where it is struck and bowed: which modes speak |
| **SIZZLE** | a rattle that follows the ring |
| **SING** | the mode the bow favours |
| **BOW** | bow pressure: from a whisper to a scream |

The mode frequencies follow measurements and theory: rectangular-bar modes and the makers' harmonic partials for the gamelan bars (Sethares), 1 : 2 : 3 for handpan and pan, 1, 2.9, 5.7, 9.1 for the bowl (Inacio, Henrique and Antunes), a clamped bar for the tongue drum, a thin plate's dense series for the cymbal, the Bessel-function modes of a membrane for the drum.

# SONG: presets, chord memory and key




| | | |
|---|---|---|
| **1** | d-pad | choose a preset slot or a setting |
| **2** | A | load the preset (your Wi-Fi setting stays) |
| **3** | X | save the sound and rhythms to the slot |
| **4** | R2 + Y | clear the slot, or the chord memory |

Ten preset slots keep everything: voices, rhythms, scale, effects, the chord memory and PLAY MODE.

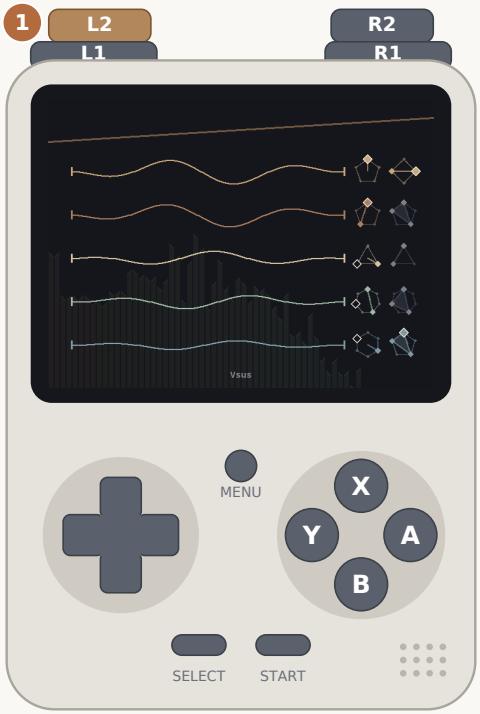
## Chord memory

Set PROG to RECORD and every chord change (SELECT or CHANGES) is kept, up to 16. PLAY steps through them round and round; FREE lets the algorithm choose again.

## Playing with others

KEY LOCK holds the root, scale and MODE (the mode chords come from in EQUAL) and tunes chords exactly. ATTACK and RELEASE set how the drone fades in and out with START.

# The quiet view and the projection




Tap L2 on PLAY for the quiet view: the five strings across the screen with their rings, and only the chord in words. Tap again to come back.

| | | |
|---|---|---|
| 1 | tap L2 | the quiet view on / off |




Behind every page a faint projection glows, like the light across the Jaffe house's ceiling: the sound's spectrum as siding boards, low notes on the left, high on the right, each pair's tops sloping into a roof valley. With binaural on, two projector beams fall from the top corners and trade left and right at the beat.

| | | |
|---|---|---|
| 1 | START + Y | projection SOFT, BRIGHT or OFF (remembered) |

# Binaural, ombak and tunings

On MIX, BEAT sets a binaural beat from 0.1 to 40 Hz: the voices below about 1 kHz are shifted down in the left ear and up in the right, so every low partial beats between the ears. DEPTH sets how much. MODE:

| | |
|---|---|
| **FREE** | the beat as set |
| **TEMPO** | the beat moves to a power-of-two multiple of the tempo |
| **A>T 10M** | a journey from alpha (10 Hz) down to theta (6 Hz) over ten minutes |
| **T>D 15M** | from theta (6 Hz) down to delta (2.5 Hz) over fifteen minutes |
| **OMBAK** | the beat for both ears, like a Balinese gamelan's paired instruments: the low end swells and fades, and it works on the speaker |

## Settings to start from

| | |
|---|---|
| **Meditation** | BEAT 6 MEDITATE, headphones; or MODE A>T 10M to drift down from alpha |
| **Sleep** | MODE T>D 15M, headphones, a long ATTACK and RELEASE |
| **Gamelan shimmer** | scale TIRUS or BEGBEG, MODE OMBAK, BEAT 7 Hz, on the speaker |

The research on binaural beats is limited and mixed; these are the settings most used, not a promise of an effect.

## Tunings

COMPOSE, or R2 + ◄ / ► on PLAY, chooses the scale: EQUAL (with its modes), just major and minor, the harmonic series, just pentatonics, the ragas YAMAN, BHAIRAV, KAFI, BHUPALI and MALKAUNS, the gamelan tunings SLENDRO, PELOG and SELISIR, and the Balinese gong kebyar tunings BEGBEG, SEDANG and TIRUS (Toth's profiles, as reproduced by Vitale and Sethares). Each scale has its own way of moving through chords.

# Quick reference




| | |
|---|---|
| START | start / stop the drone |
| START + L1/R1 | previous / next page |
| START + A | PLAY MODE: RINGS / LATCH |
| **LATCH** B / YX | harmonic (held) / bow attack |
| START + Y | projection SOFT / BRIGHT / OFF |
| START + SELECT | the log: OFF / BASIC / DETAIL |
| SELECT | next chord |
| L1 | tap: strike; hold: pluck (wind: hard attack / tongue); LATCH: on / let go |
| R1 | bow while held (wind: blow); LATCH: bow on / let go |
| L1 + R1 | harmonic (wind: overblow), either first |
| **tap** L2 | quiet view |
| L2 + **...** | shape the ring, mute (A) |
| R2 + **...** | scale, tone, chance, body |
| MENU | leave |

## The log

START + SELECT cycles the log through OFF, BASIC and DETAIL. It is written to App/Jaffedrone/jaffedrone_log.txt: how hard the audio is working, any dropouts, the levels, each voice's tuning (DETAIL) and what was played, with suggestions when something could be better. If you hear crackles, set JD_BUFFER=2048 in App/Jaffedrone/launch.sh.